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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 5: Climate Change: Indicators and Impacts

Prof. Bhola Ram Gurjar

Dept. of Civil Engineering



A Few Questions ?



- Is climate really changing over the years?
- What are indicators and impacts of climate change?
- If so, how it is linked with the present model of development?
- How can we address climate change and develop sustainable development strategies?

What is Climate Change ?



- Change in usual weather found in a place.

  Ex. Changes in how much rain a place gets in a year, Change in a place's usual temperature for a month or season.

- Climate change is also change in Earth's Climate.

  Ex. Change in Earth's usual temperature, Change in the places where rain and snow falls on Earth.

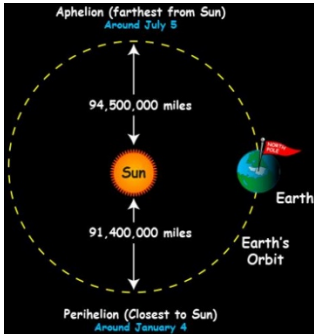
- Weather can change in a few hours whereas Climate takes hundreds or even millions of years to change.

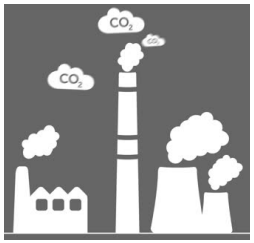
Source: (NASA, 2014)

How is Climate Changing ?

- Natural reasons (Not Controllable)
  - Earth's distance from the sun
  - Oceans, Volcano eruptions etc.
- Humans (Controllable)
  - Anthropogenic activities.
  - GHG emissions.







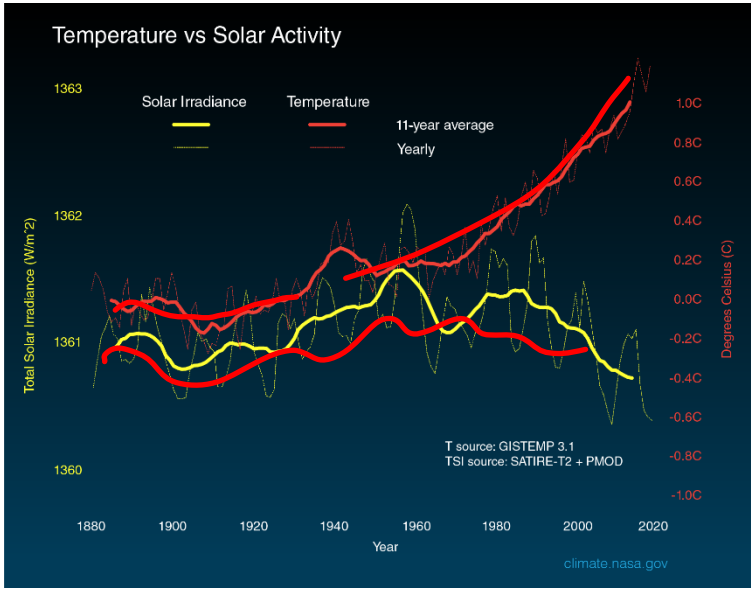


Source: (NASA, 2014)

Is Global warming manmade ?

- The amount of solar energy received by the Earth has followed the Sun's natural 11-year cycle of small ups and downs with no net increase since the 1950s.

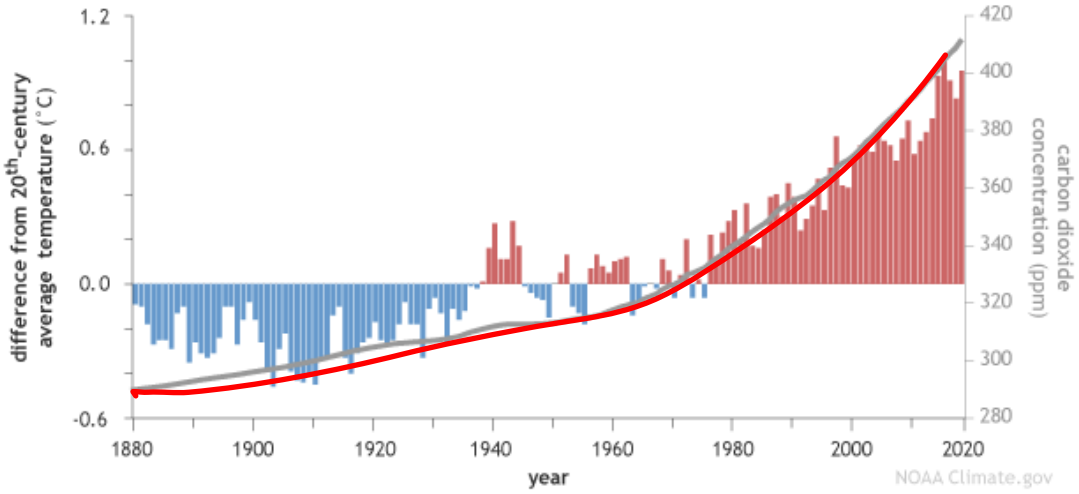
- Over the same period, global temperature has risen markedly.



It is therefore extremely unlikely that the Sun has caused the observed global temperature warming trend over the past half-century.

Source: (NASA, 2020. https://climate.nasa.gov/climate_resources/189/graphic-temperature-vs-solar-activity/)

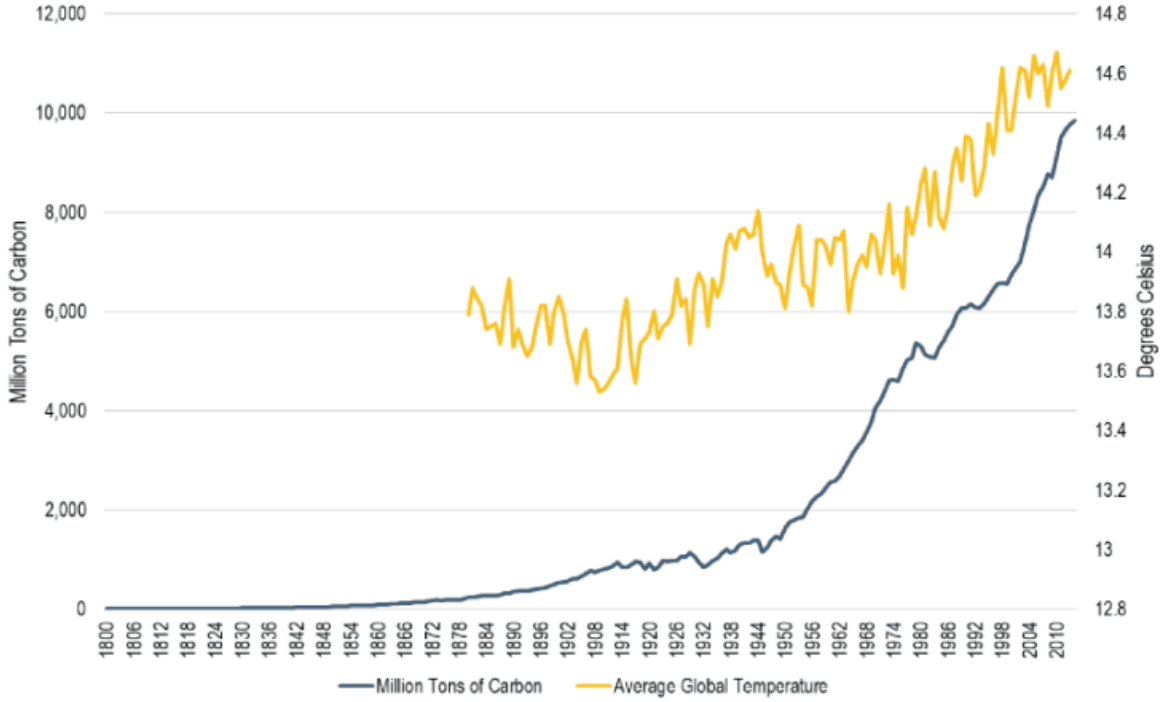
Atmospheric CO_2 and Earth's Surface temperature



- Yearly temperature compared to the twentieth-century average from 1880–2019 (red and blue bars)
- Atmospheric carbon dioxide concentrations from 1880-2019 (grey line)

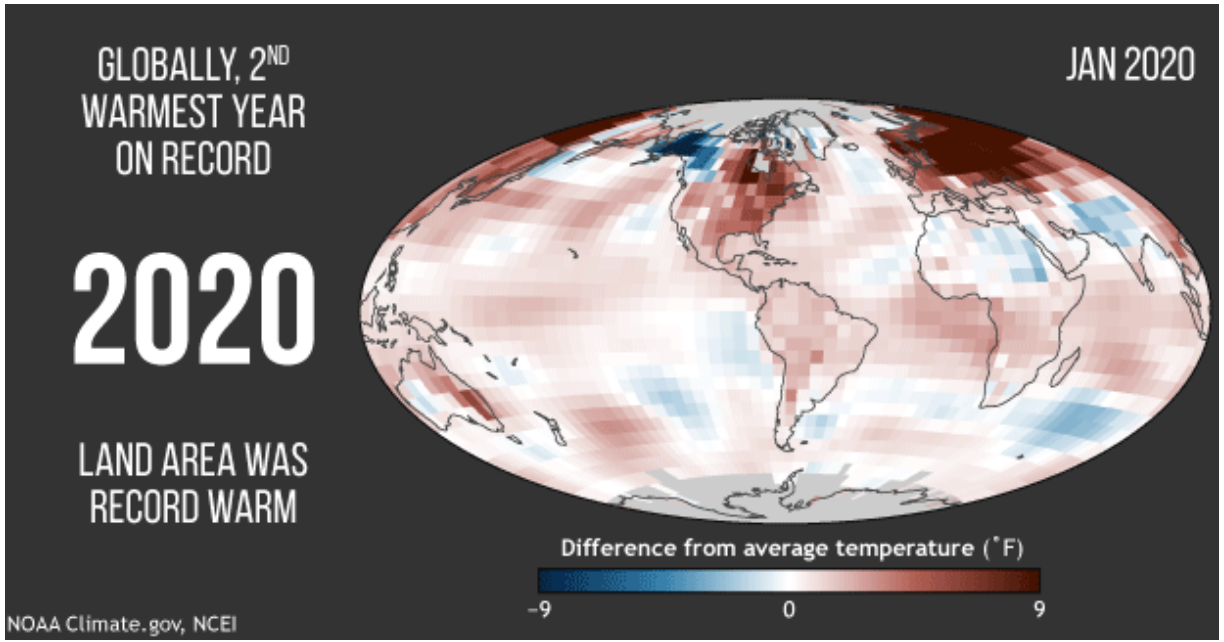
Source: (https://www.climate.gov/news-features/climate-qa/if-carbon-dioxide-hits-new-high-every-year-why-isn%E2%80%99t-every-year-hotter-last)

Average Global Temperature and World Carbon Emissions From Fossil Fuel Burning, (in millions of tons) 1800-2014



Source: (National Oceanic and Atmospheric Administration, Earth System Research Laboratory, Global Monitoring Division)

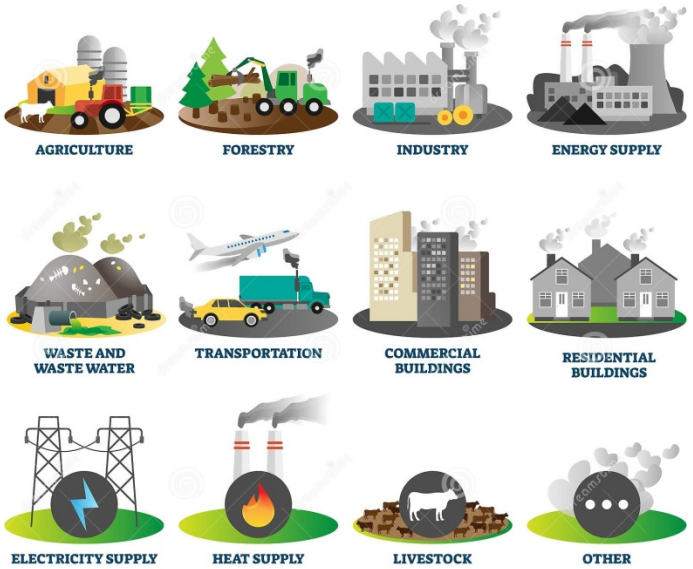
2020: 2nd Warmest Year on Record



- 2016 was the warmest year on record and 2019 the 3rd warmest.
- World's 7 warmest years have all occurred since 2014.
- World's 10 warmest years have all occurred since 2005.

Climate Change Drivers

Greenhouse Gases (GHG)



- Most significant driver of observed climate change since the mid-20th century
- GHGs resulting from human activities- Electricity, Transportation, Agriculture, Commercial & Residential
- When comparing emissions of different gases, the concept called “global warming potential” is used to convert amounts of other gases into carbon dioxide equivalents.

Source: (USEPA, https://www.epa.gov/climate-indicators/greenhouse-gases)

Climate Change Drivers: Major GHGs and its Characteristics

Greenhouse gas	Sources	Average lifetime	100-year GWP
Carbon dioxide	Burning of fossil fuels, solid waste, and trees and wood products, Deforestation and soil degradation	Varies*	1
Methane	Production and transport of oil, natural gas & coal, Livestock and agriculture, waste decomposition	12.4 years**	28-36
Nitrous oxide	Agricultural and industrial activities, Combustion of fossil fuels and solid waste.	121 years**	265-298
Fluorinated gases	Industrial processes and commercial and household uses	Few weeks to 1000 years	Varies upon the gas)

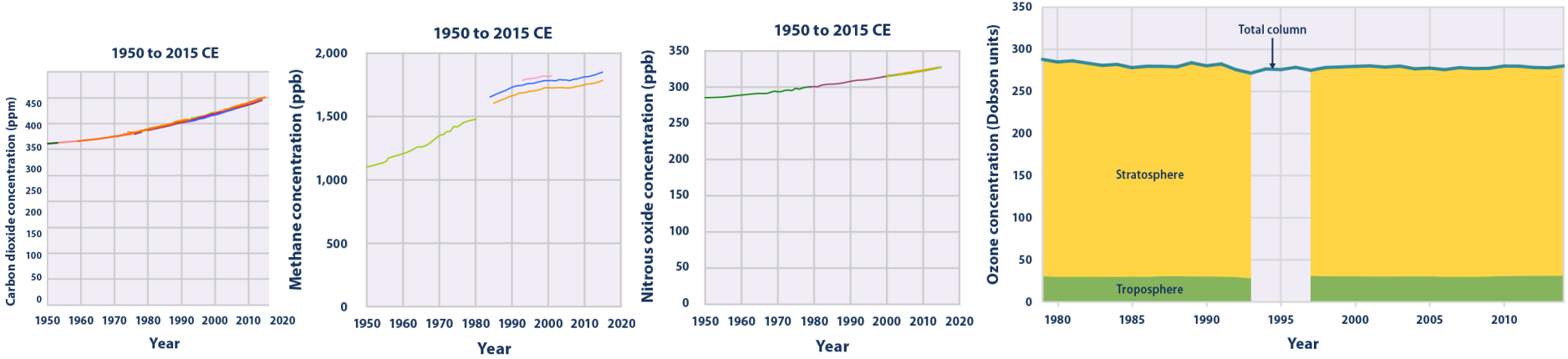
*CO_2 may get absorbed quickly (ocean surface) or may remain even thousands of years

**Perturbation lifetimes

Source: (USEPA, https://www.epa.gov/climate-indicators/greenhouse-gases)

Climate Change Drivers: GHGs

Global Atmospheric concentrations of major GHGs

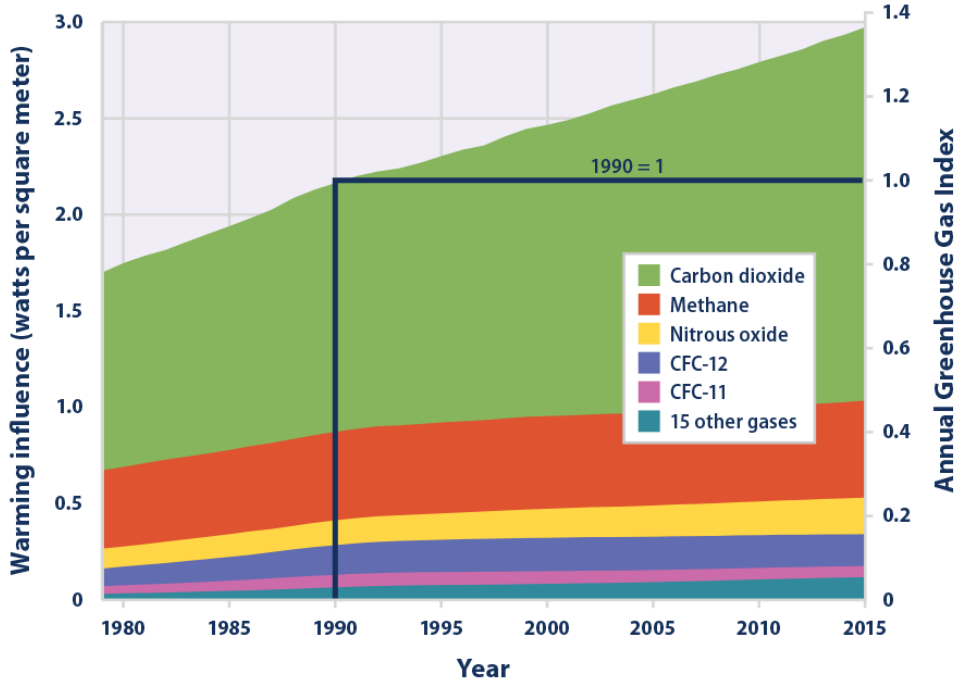


Global Atmospheric concentrations of major GHGs (CO_2, CH_4, N_2O & O_3 from 1950 – 2015)

Source: (USEPA, https://www.epa.gov/climate-indicators/climate-change-indicators-atmospheric-concentrations-greenhouse-gases)

Climate Change Drivers: GHGs

Radiative Forcing caused by major long-lived GHGs

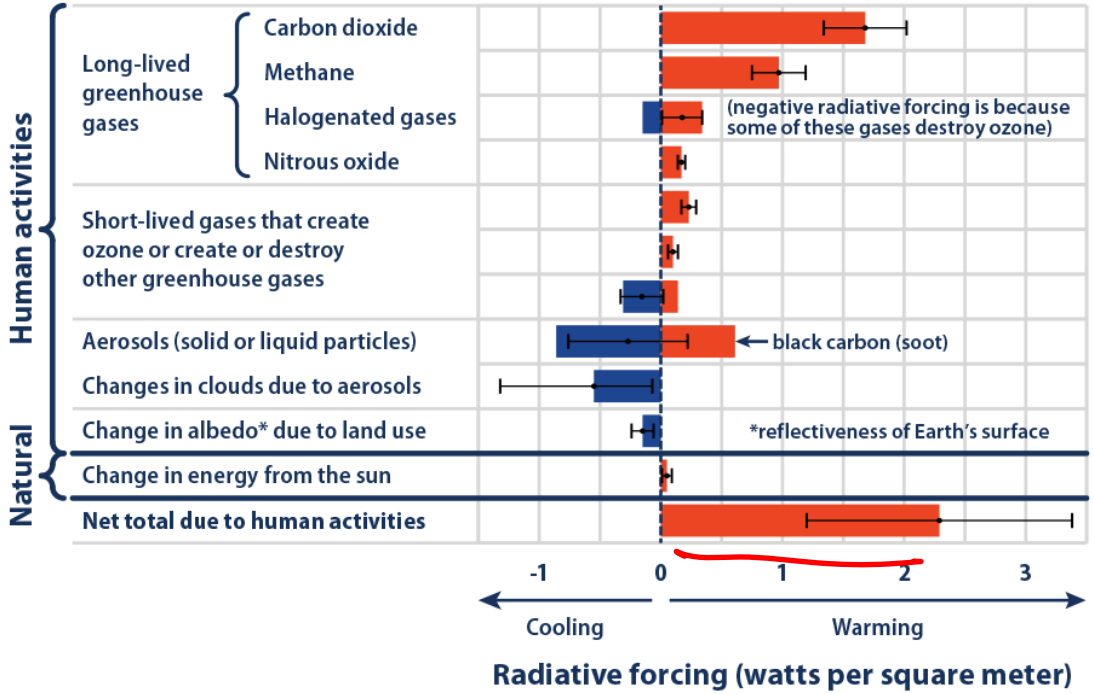


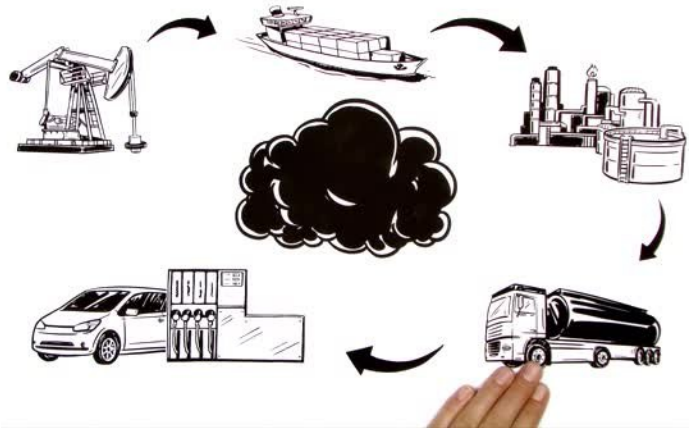
Source: (USEPA, https://www.epa.gov/climate-indicators/climate-change-indicators-climate-forcing)

Radiative forcing" is the heating effect caused by greenhouse gases in the atmosphere.

Climate Change Drivers

Radiative Forcing caused by Human activities





Source: (USEPA, https://www.epa.gov/climate-indicators/climate-change-indicators-climate-forcing)

Climate Change Indicators

Weather and Climate

- Rising global average temperature is associated with widespread changes in weather patterns.
- Extreme weather events such as heat waves and large storms are likely to become more frequent or more intense with human-induced climate change.
- Long-term changes in climate can directly or indirectly affect many aspects of society.
- Extreme variations in weather are also a threat to society- Illness to Death



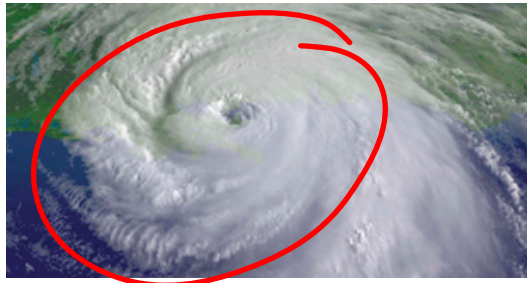
Source: (USEPA, https://www.epa.gov/climate-indicators/weather-climate)

Climate Change Indicators

Weather and Climate

- Increased precipitation can replenish water supplies and support agriculture (Positive effect)

- Intense storms can damage property, cause loss of life and population displacement, and temporarily disrupt essential services such as transportation, telecommunications, energy, and water supplies (Negative effect)







Source: (USEPA, https://www.epa.gov/climate-indicators/weather-climate)

Climate Change Indicators: Oceans

- Covers about 70% of Earth's surface
- Changes in weather can influence the weather from local to global scales and can alter many properties of Oceans
- As GHGs absorb more energy from the sun, Oceans absorbs more heat, resulting in an increase in sea surface temperatures and rising sea level.
- Changes in ocean temperatures and currents brought about by climate change will lead to alterations in climate patterns around the world.



Source: (USEPA, https://www.epa.gov/climate-indicators/oceans)

Climate Change Indicators: Oceans

- Oceans help reduce climate change by storing large amounts of carbon dioxide (Positive effect)



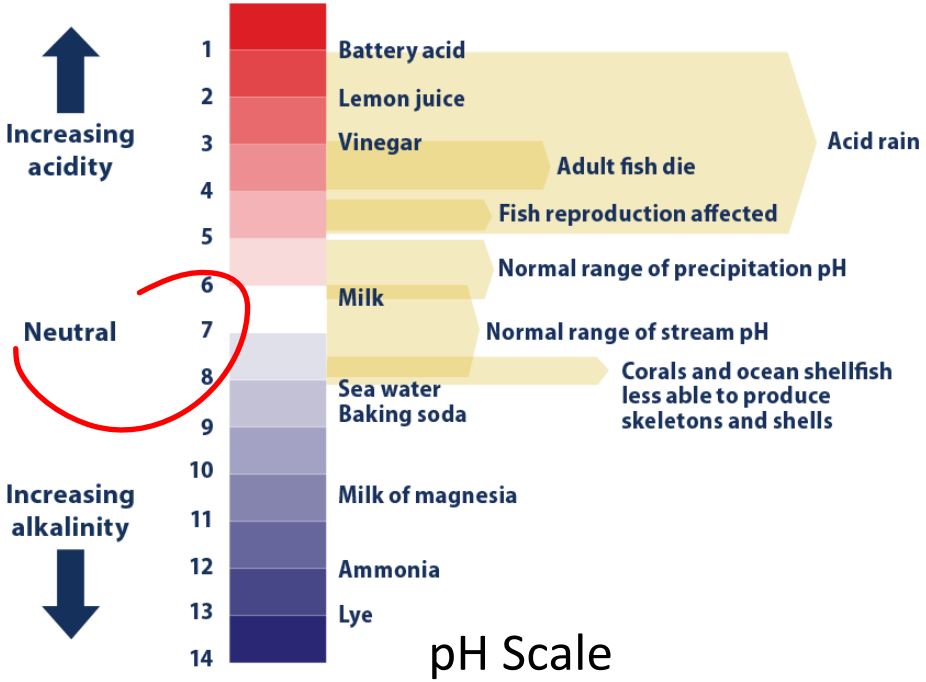
- Increasing levels of dissolved carbon are changing the chemistry of seawater and making it more acidic (Negative effect)

- Could substantially alter the biodiversity and productivity of ocean ecosystems (Negative effect)
  - Ex. Increased ocean acidity makes it more difficult for certain organisms, such as corals and shellfish, to build their skeletons and shells.

Source: (USEPA, https://www.epa.gov/climate-indicators/oceans)

Climate Change Indicators: Oceans

Ocean Acidity

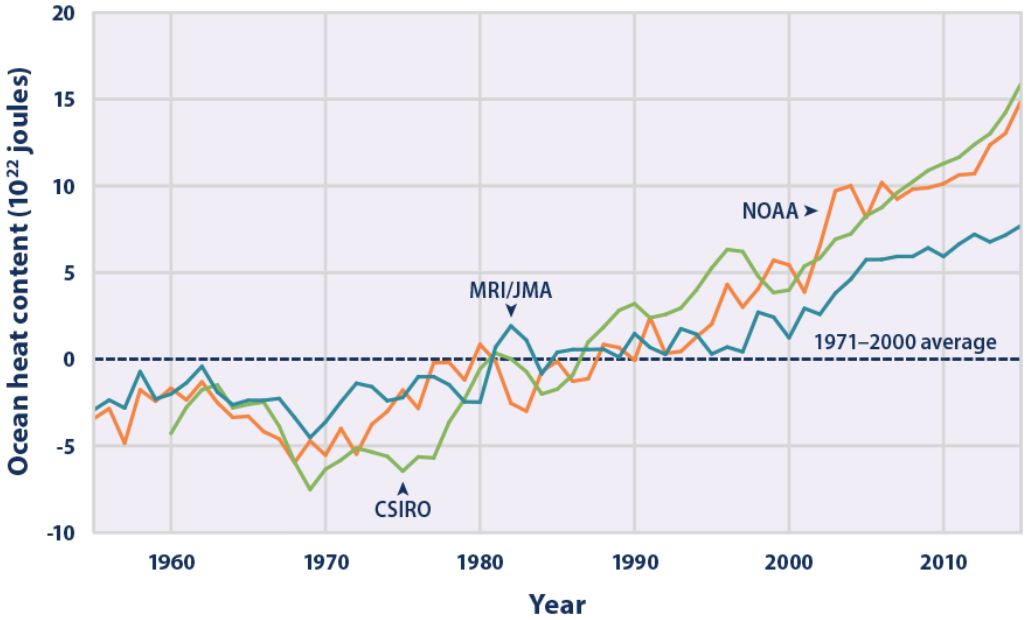


- Measurements over the last few decades have demonstrated that ocean carbon dioxide levels have risen in response to increased carbon dioxide in the atmosphere, leading to an increase in acidity (a decrease in pH)

Source: (USEPA, https://www.epa.gov/climate-indicators/climate-change-indicators-ocean-acidity)

Climate Change Indicators: Oceans

Ocean Heat

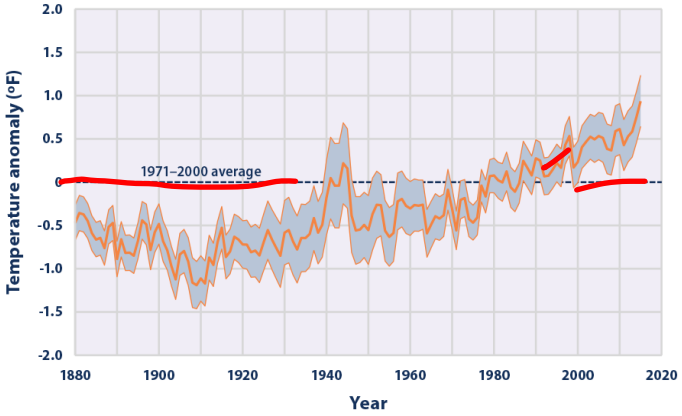


- Graph shows Ocean heat content between 1955 and 2015
- Measured in joules and compared against the 1971–2000 average, which is set at zero for reference

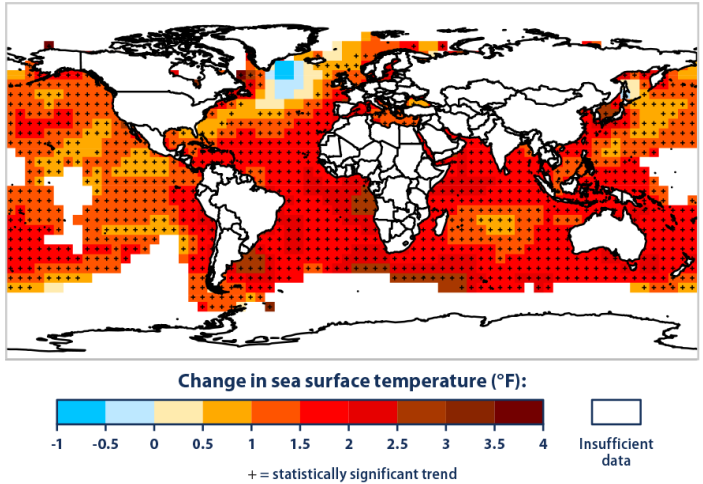
Source: (USEPA, https://www.epa.gov/climate-indicators/climate-change-indicators-ocean-heat)

Climate Change Indicators: Oceans

Sea Surface Temperature



Average Global Sea Surface Temperature, 1880–2015



Change in Sea Surface Temperature, 1901–2015

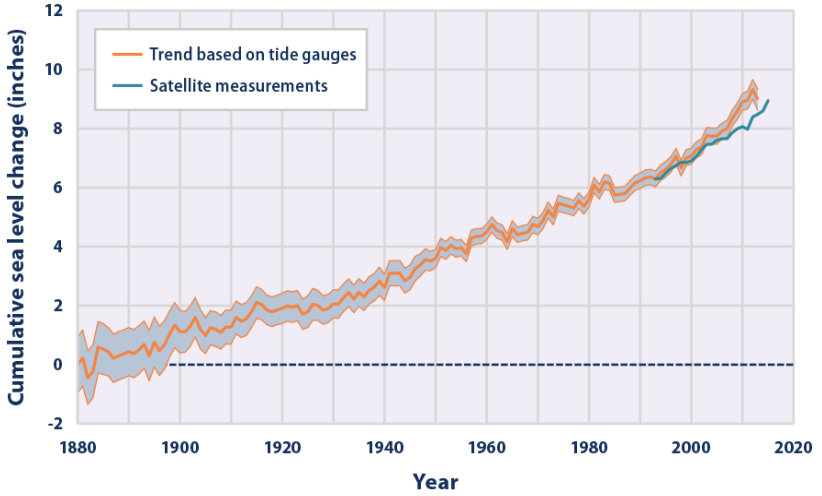
- Sea surface temperature increased during the 20th century
- From 1901 through 2015, temperature rose at an average rate of 0.13°F per decade

Source: (USEPA, https://www.epa.gov/climate-indicators/climate-change-indicators-sea-surface-temperature)

Climate Change Indicators: Oceans

Sea Level

- The Global average sea level rose throughout the 20th century, and the rate of change has accelerated in recent decades.
- (Absolute sea level has risen at an average rate of 0.06 inches per year from 1880 to 2013)
- Since 1993, the average sea level has risen at a rate of 0.11 to 0.14 inches per year—roughly twice as fast as the long-term trend.



Global Average Absolute Sea Level Change, 1880–2015

Source: (USEPA, https://www.epa.gov/climate-indicators/climate-change-indicators-sea-level)

Climate Change Indicators: Snow and Ice

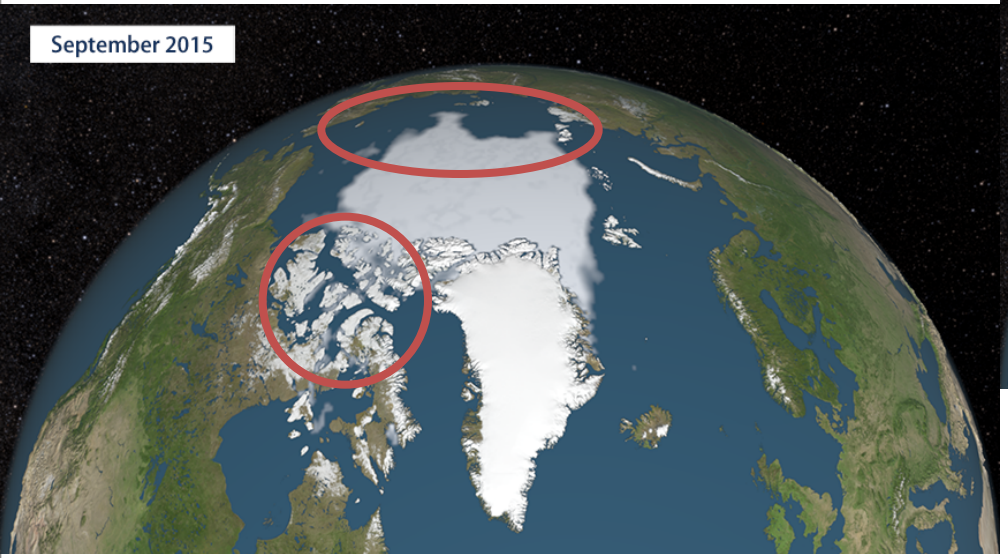
- Earth's surface contains many forms of snow and ice including sea, lake, and river ice; snow cover; glaciers, ice caps, and ice sheets; and frozen ground.
- Climate change can dramatically alter the Earth's snow- and ice-covered areas in response to relatively minor changes in temperature.
- Reduced snow and ice can influence air temperatures, sea level, ocean currents, and storm patterns

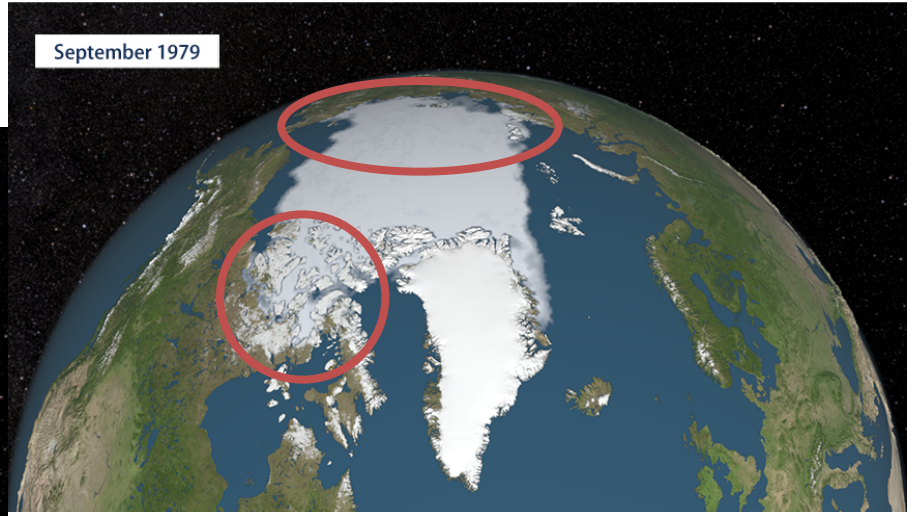


Source: (USEPA, https://www.epa.gov/climate-indicators/snow-ice)

Climate Change Indicators:

Snow and Ice: Arctic Sea Ice

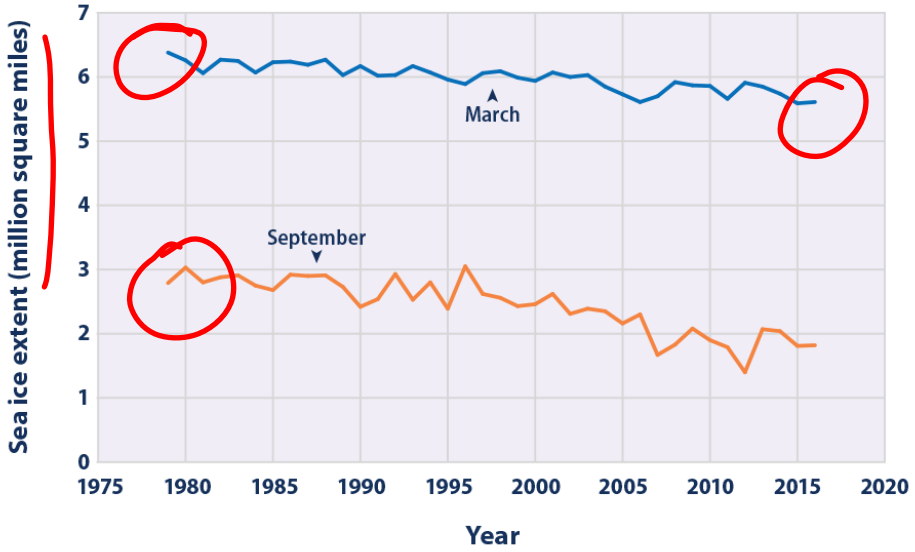




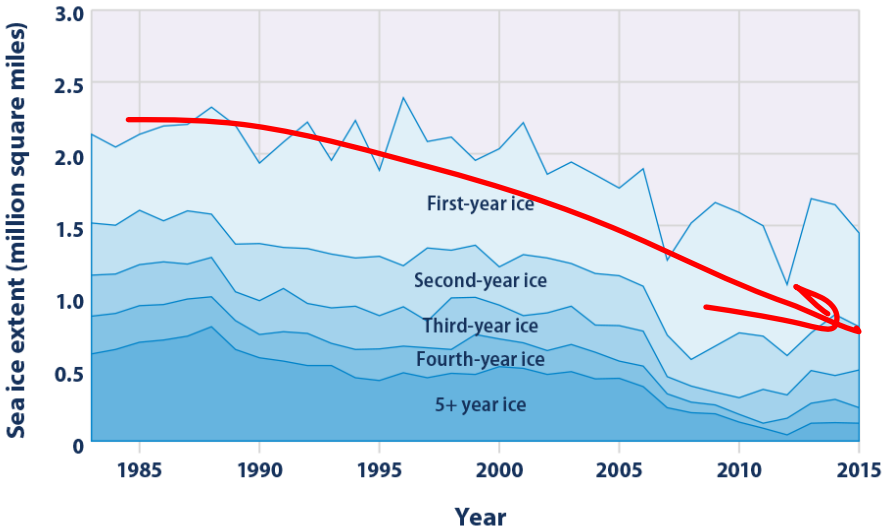
Dwindling Arctic Sea Ice (NASA, 2016)

Climate Change Indicators:

Snow and Ice: Arctic Sea Ice



March (Max.) and September (Min.) Monthly Average Arctic Sea Ice Extent, 1979–2016

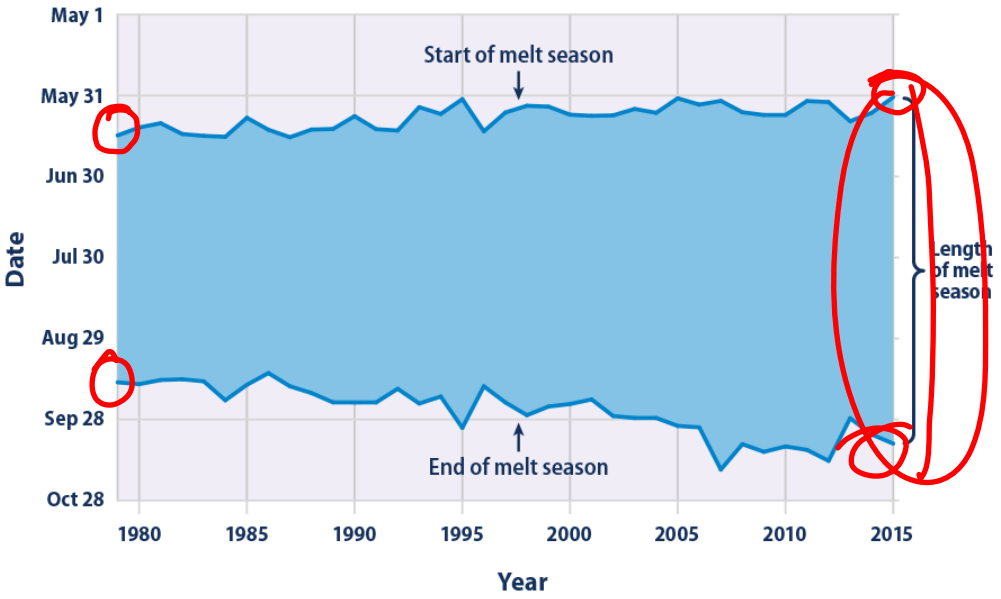


Age of Arctic Sea Ice at Minimum September Week, 1983–2015

Source: (USEPA, https://www.epa.gov/climate-indicators/climate-change-indicators-arctic-sea-ice)

Climate Change Indicators:

Snow and Ice: Arctic Sea Ice



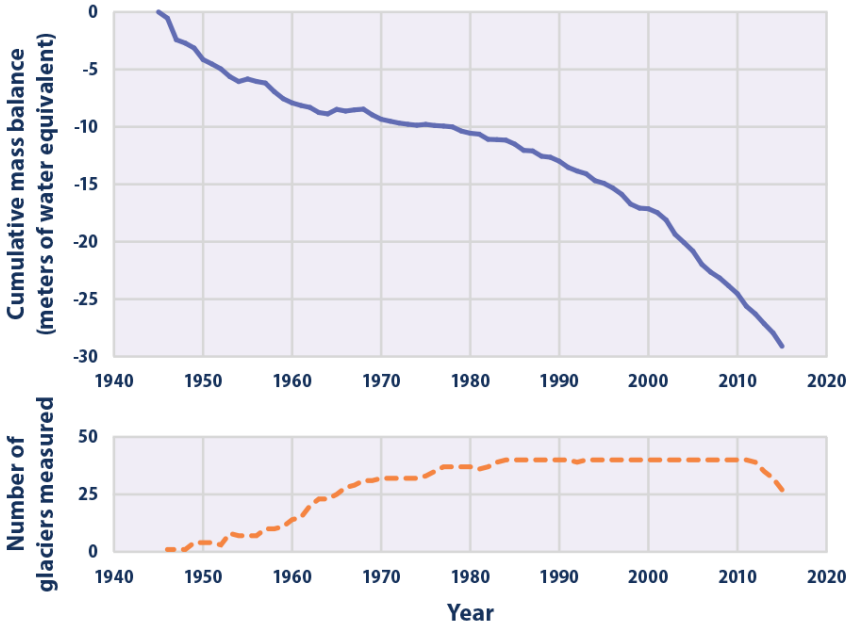
Arctic Sea Ice Melt Season, 1979–2015

- The shaded band spans from the date when ice begins to melt consistently until the date when it begins to refreeze.
- The Length of seasonal meltdown has increased over the years.

Source: (USEPA, https://www.epa.gov/climate-indicators/climate-change-indicators-arctic-sea-ice)

Climate Change Indicators:

Snow and Ice: Glaciers



Average Cumulative Mass Balance of Glaciers Worldwide, 1945–2015

- Glaciers worldwide have been losing mass since at least the 1970s, contributed in sea level rise significantly.
- The rate at which glaciers are losing mass appears to have accelerated over roughly the last decade.

Source: (USEPA, https://www.epa.gov/climate-indicators/climate-change-indicators-glaciers)

Climate Change Impacts: Health and Society

- Changes in the Earth's climate can affect public health, agriculture, water supplies, energy production and use, land use and development, and recreation.
- Impacts on human health are complex and mostly indirect, and dependent on multiple societal and environmental factors (including how people choose to respond to these impacts)
- The development of appropriate health-related climate indicators is challenging and still emerging.



Source: (USEPA, https://www.epa.gov/climate-indicators/health-society)

Climate Change Impacts: Health and Society

- Changes in climate affect the average weather conditions

Ex. Warmer average temperatures will likely lead to hotter days and more frequent and longer heat waves, which could increase the number of heat-related illnesses and deaths.



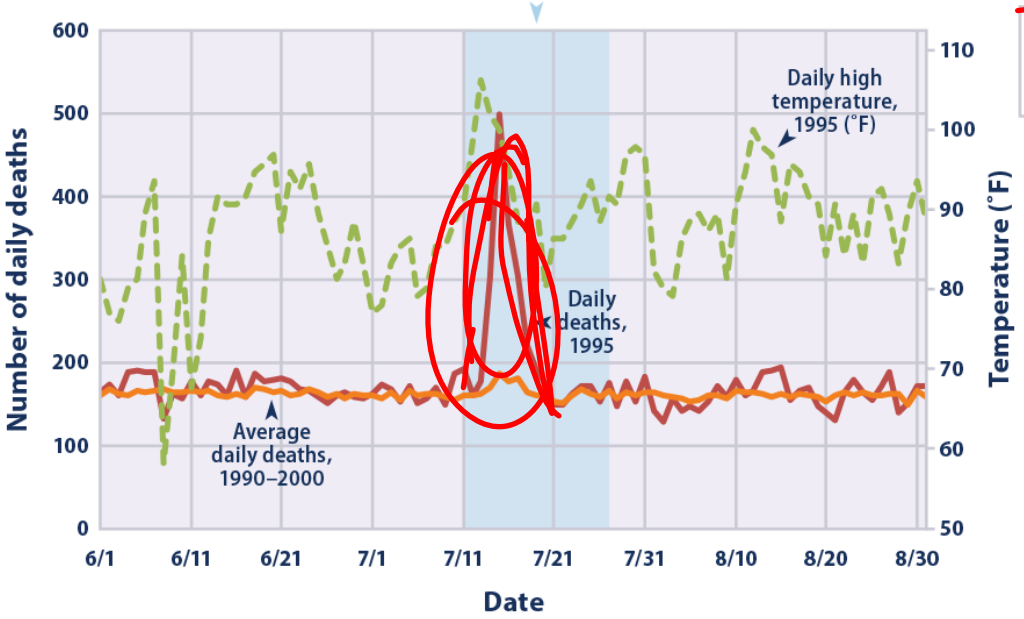
- Extreme weather events, such as storms, could increase the risk of dangerous flooding, high winds, and other direct threats to people and property.

- Warmer temperatures could also reduce air quality, spreading diseases

Source: (USEPA, https://www.epa.gov/climate-indicators/health-society)

Climate Change Impacts: Health and Society

The Chicago Heat Wave, Cook County, July 11-27, 1995



Heat-Related Deaths During the 1995 Chicago Heat Wave

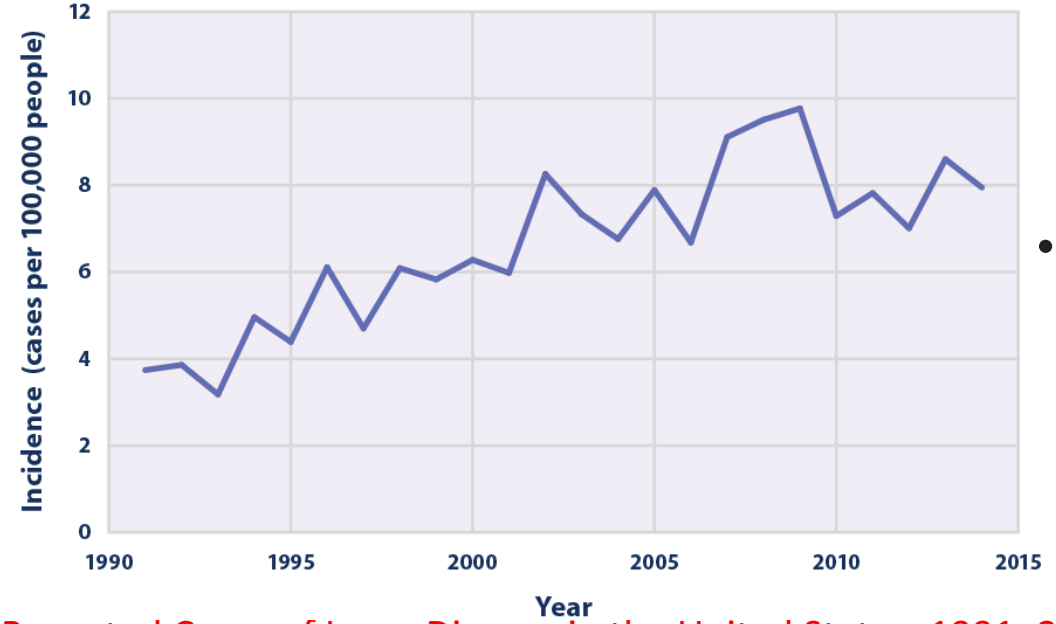
Cook County, July 11–27, 1995:
Excess deaths compared with this time period during an average year: about 700
Deaths classified as "heat-related" on death certificates (not shown here): 465

- Studies of the 1995 heat wave, Chicago suggests that there may have been hundreds more deaths than were actually reported as "heat-related" on death certificates.

Source: (USEPA, https://www.epa.gov/climate-indicators/climate-change-indicators-heat-related-deaths)

Climate Change Impacts: Health and Society

Lyme Disease (a bacterial disease)



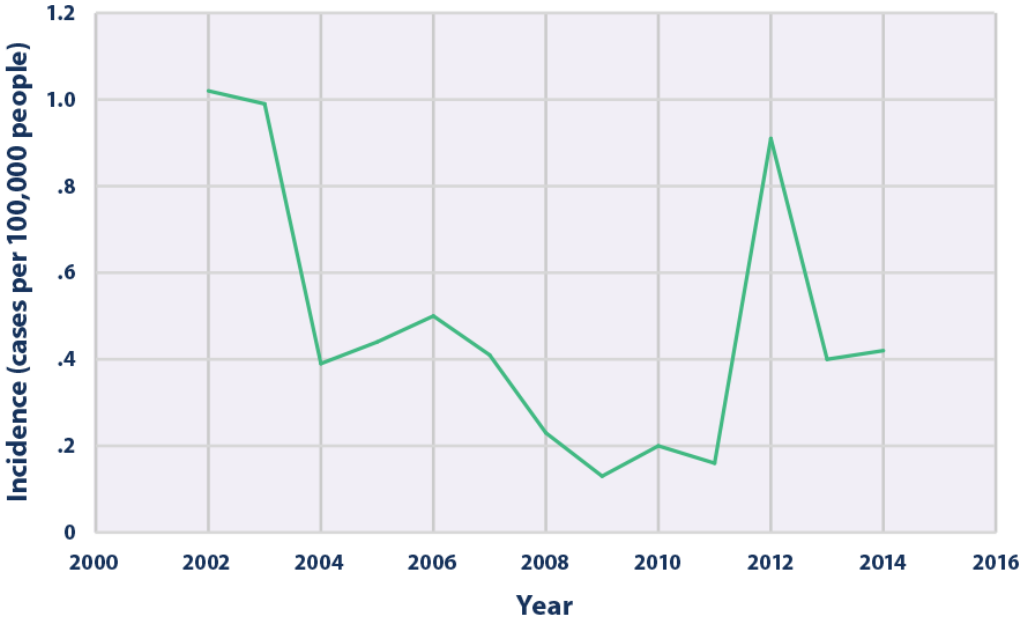
Reported Cases of Lyme Disease in the United States, 1991–2014

- The incidence of Lyme disease in the U.S has approximately doubled from 1991 to 2014.
- 3.74 reported cases per 100,000 people in 1991 to 7.95 reported cases per 100,000 people in 2014.

Source: (USEPA, https://www.epa.gov/climate-indicators/climate-change-indicators-lyme-disease)

Climate Change Impacts: Health and Society

West Nile Virus



Reported Neuroinvasive Cases of West Nile Virus Disease in the United States, 2002-2014

- Neuroinvasive cases occur for less than 1% of people infected with West Nile virus
- Neuroinvasive cases affects the brain or cause neurologic dysfunction.

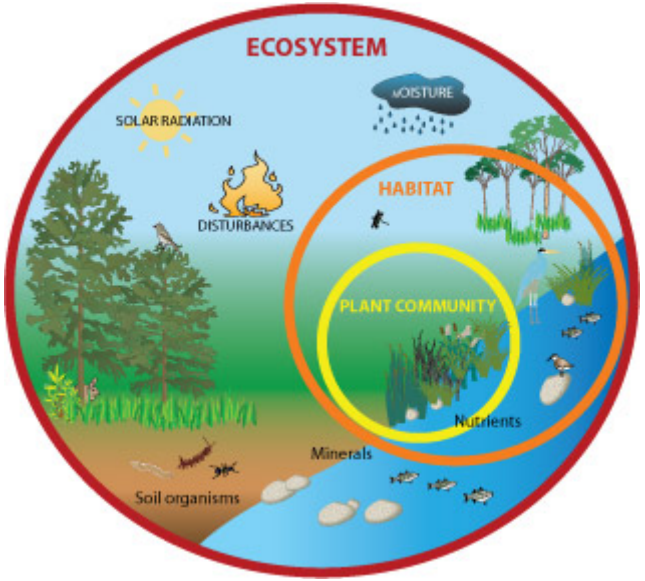
Source: (USEPA, https://www.epa.gov/climate-indicators/climate-change-indicators-west-nile-virus)

Climate Change Impacts: Ecosystems

- Ecosystems provide humans with food, clean water, and a variety of other services that can be affected by climate change.

- Changes in the Earth's climate can affect ecosystems by altering the water cycle, habitats, animal behaviour. Ex. nesting and migration patterns, the timing of natural processes such as flower blooms.

- May increase the risk of harm or even extinction for some species.



Source: (USEPA, https://www.epa.gov/climate-indicators/ecosystems)

Climate Change Impacts: Ecosystems

Wildfires



- Very intense Wild fires can significantly disrupt ecosystems, damage property, put people and communities at risk, and create air pollution problems even far away from the source.

Source: (USEPA, https://www.epa.gov/climate-indicators/ecosystems)

Climate Change Impacts: Ecosystems

Streamflow



- Changes in temperature, precipitation, snowpack, and glaciers can affect the rate of streamflow and the timing of peak flow.

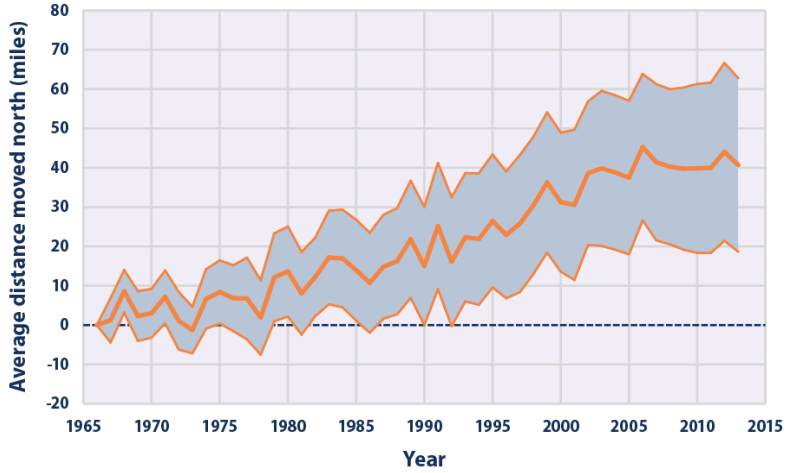
Source: (USEPA, https://www.epa.gov/climate-indicators/ecosystems)

Climate Change Impacts: Ecosystems

Bird Wintering Ranges

- Some birds shift their range or alter their migration habits to adapt to changes in temperature or other environmental conditions.

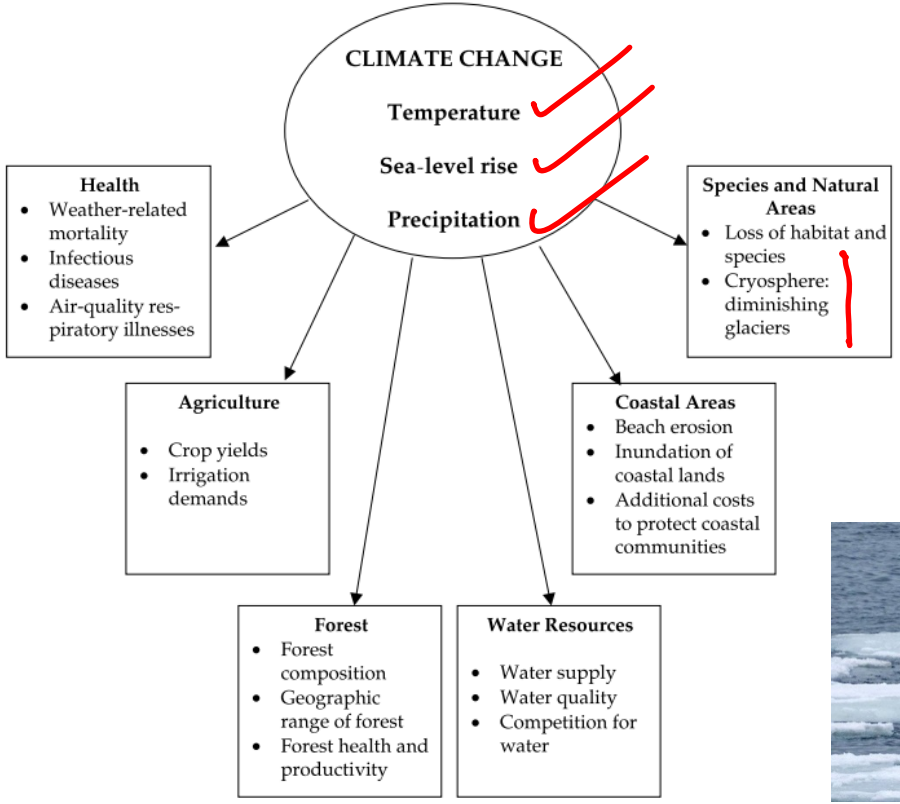
Ex. Long-term studies have found that bird species in North America have shifted their wintering grounds northward by an average of more than 40 miles since 1966, with several species shifting by hundreds of miles.



Change in Latitude of Bird Center of Abundance, 1966–2013

Source: (USEPA, https://www.epa.gov/climate-indicators/climate-change-indicators-bird-wintering-ranges)

Summary: Potential Impacts of Climate Change

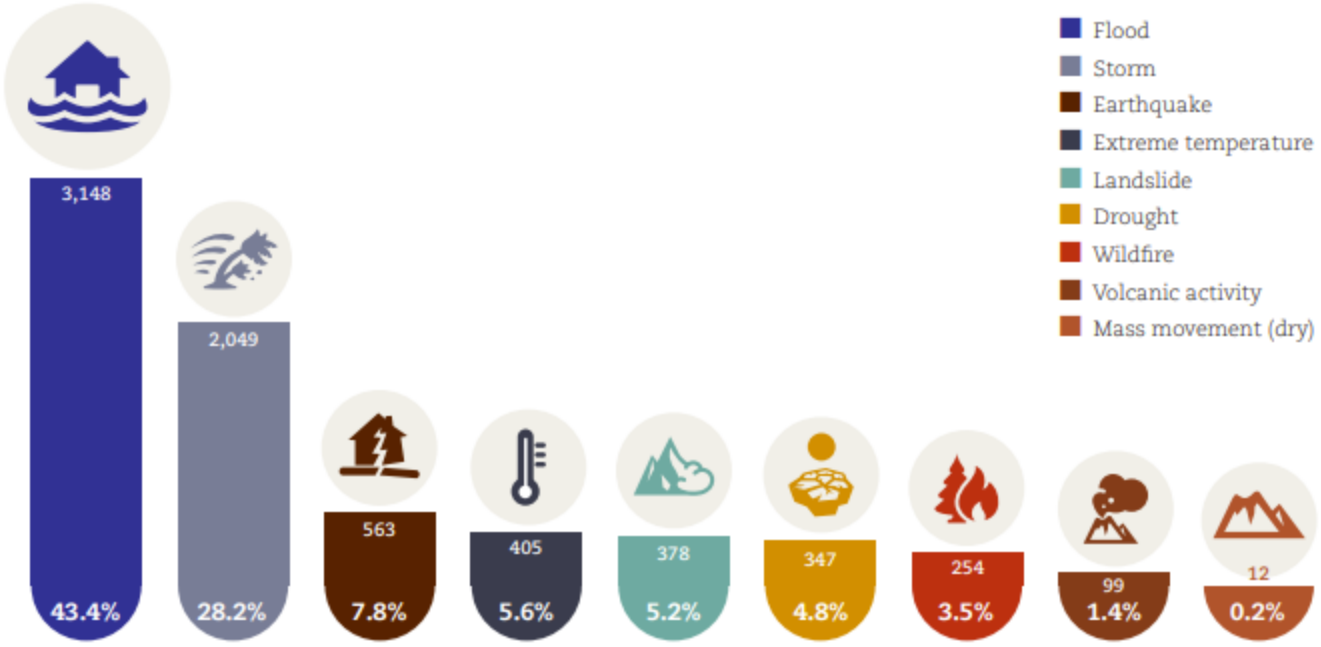






Source: (Cambridge Books online, 2010)

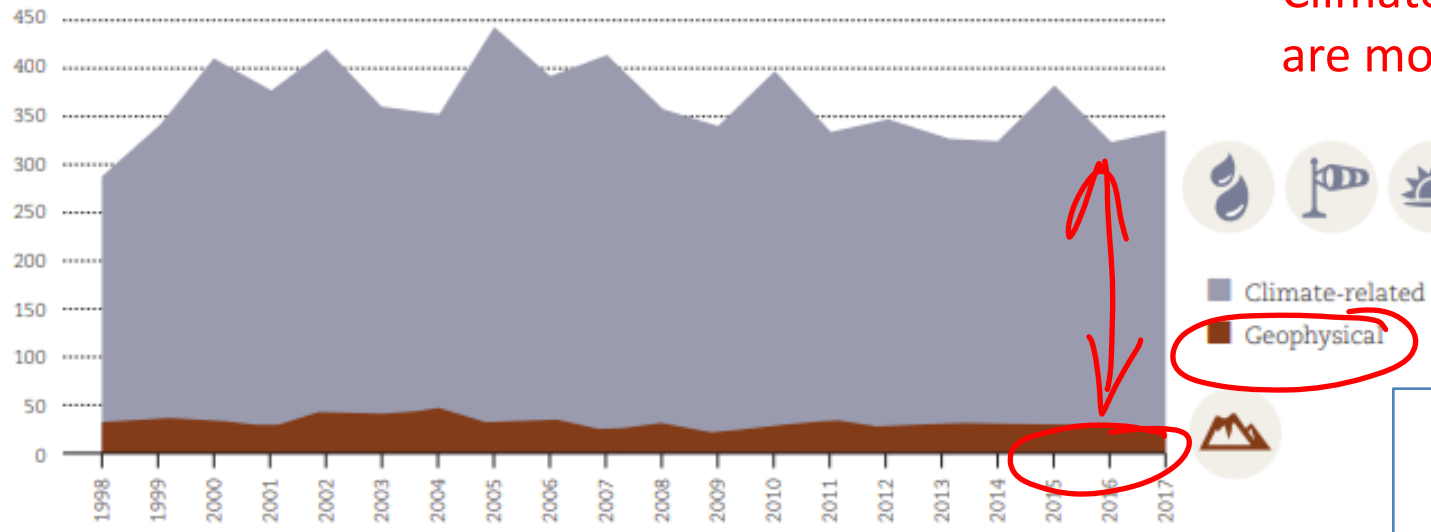
Climate Change: Worldwide Disasters (1998-2017)



Number of disasters per type (1998-2017)

Source: (UNISDR, 2017)

Climate Change: Worldwide Disasters (1998-2017)

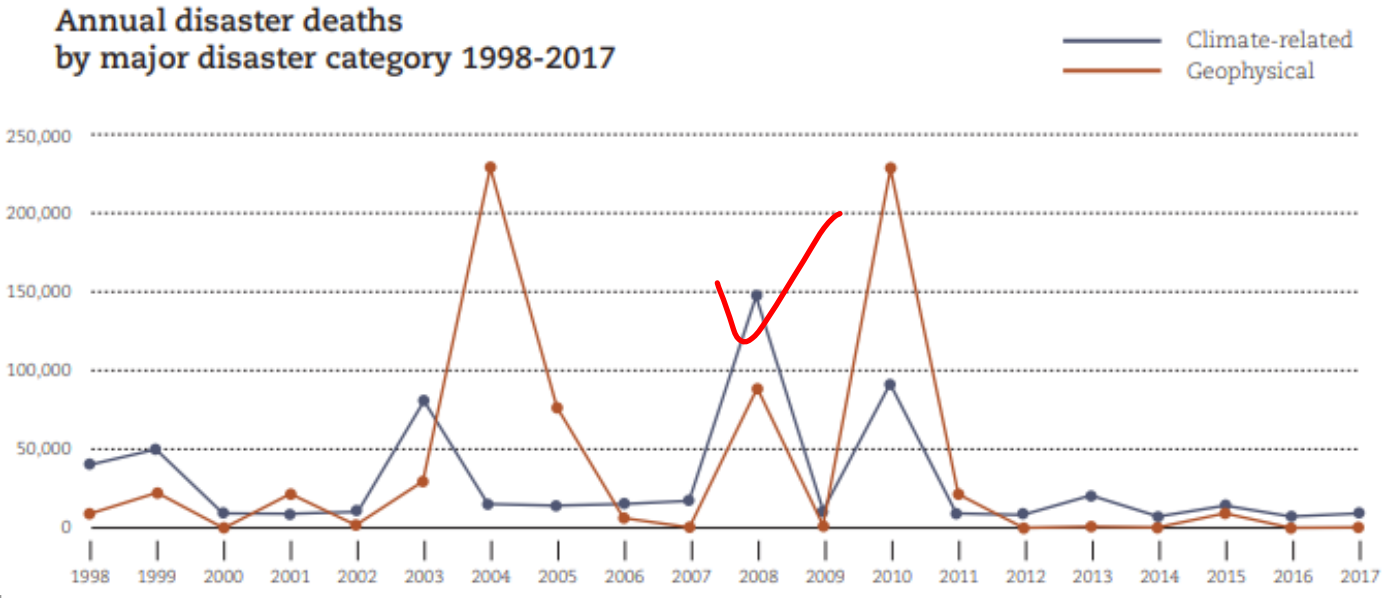


Number of disasters by major category per year 1998-2017

- Climate related disasters are more significant

Source: (UNISDR, 2017)

Climate Change: Worldwide Disasters (1998-2017)

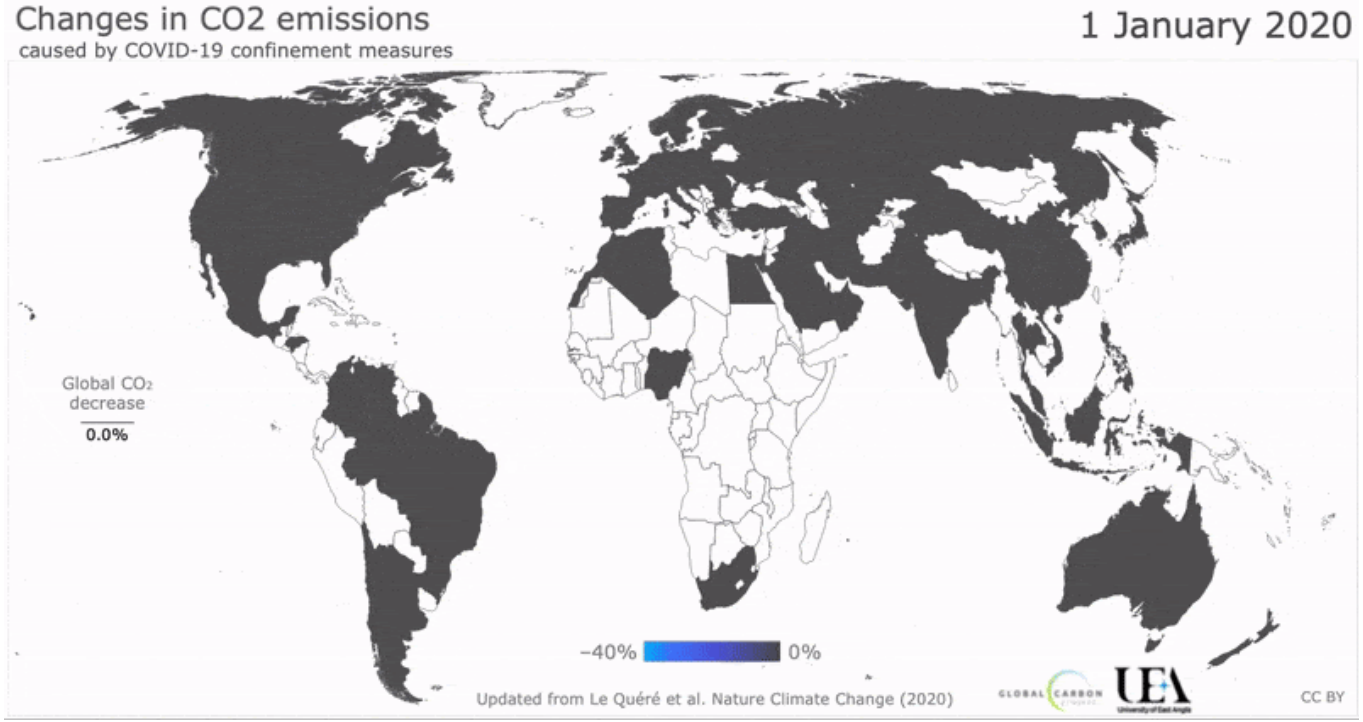


Annual deaths from disasters (1998-2017)

Climate related deaths are more in the last decade

Source: (UNISDR, 2017)

Impact of COVID-19 on CO_2 emissions



This improvement is only temporary !!!

Source: (https://www.carbonbrief.org/global-carbon-project-coronavirus-causes-record-fall-in-fossil-fuel-emissions-in-2020)

Key facts

- Climate change affects the social and environmental determinants of health – clean air, safe drinking water, sufficient food and secure shelter.

- Between 2030 and 2050, climate change is expected to cause approximately 250,000 additional deaths per year, from malnutrition, malaria, diarrhoea and heat stress.

- The direct damage costs to health is estimated to be between USD 2-4 billion/year by 2030.

Take urgent action to combat climate change and its impacts

Reducing emissions of GHGs through better transport, food and energy-use choices can result in improved health, particularly through reduced air pollution.

Source: (WHO, 2018)

Conclusion

- Climate change is affecting every country on every continent.
- It is disrupting national economies and affecting lives.
- Weather patterns are changing, sea levels are rising, and weather events are becoming more extreme.
- Although travel bans and economic slowdowns resulting from the COVID-19 pandemic may reduce the GHG emissions, but this improvement is only temporary.
- Climate change is not on pause.

Switch to Sustainable measures

Sustainable development, that essentially includes Sustainable Transportation Systems, is the most feasible mitigation measure to Climate change.



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Thank You